

SECTION 16060

GROUNDING AND BONDING

PART 1 GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes grounding of electrical systems and equipment. Grounding requirements specified in this Section may be supplemented by special requirements of systems described in other Sections.

1.3 SUBMITTALS

- A. Field Test Reports: Submit written test reports to include the following:
 - 1). Test procedures used.
 - 2). Test results that comply with requirements.
 - 3). Results of failed tests and corrective action taken to achieve test results that comply with requirements.

1.4 QUALITY ASSURANCE

- A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.
 - 1). Comply with UL 467.
- B. Comply with NFPA 70; for overhead-line construction and medium-voltage underground construction, comply with IEEE C2.
- C. Comply with NFPA 780 and UL 96 when interconnecting with lightning protection system.
- D. Comply with IEEE 80 and IEEE 837 for substation grounding.

PART 2 PRODUCTS

2.1 MANUFACTURERS

- A. Available Manufacturers: Subject to compliance with requirements, manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
- 1). Grounding Conductors, Cables, Connectors, and Rods:
 - a). Chance/Hubbell
 - b). Copperweld Corp.
 - c). Erico Inc.; Electrical Products Group
 - d). Burndy Electrical
 - e). Harger Lightning Protection, Inc.
 - f). Heary Brothers Lightning Protection Co.
 - g). Ideal Industries, Inc.
 - h). ILSCO
 - i). Kearney/Cooper Power Systems
 - j). Korns: C. C. Korns Co.; Division of Robroy Industries
 - k). Lightning Master Corp.
 - l). Lyncole XIT Grounding
 - m). O-Z/Gedney Co.; a business of the EGS Electrical Group
 - n). Robbins Lightning, Inc.
 - o). Thomas & Betts, Electrical
- B. Similar materials shall be procured from one manufacturer to insure compatibility.

2.2 GROUNDING CONDUCTORS

- A. For insulated conductors, comply with Division 16 Section "Conductors and Cables."
- B. Material: copper.
- C. Equipment Grounding Conductors: Insulated with green-colored insulation.
- D. Isolated Ground Conductors: Insulated with green-colored insulation with yellow stripe. On feeders with isolated ground, use colored tape, alternating bands of green and yellow tape to provide a minimum of three bands of green and two bands of yellow.
- E. Grounding Electrode Conductors: Stranded cable.
- F. Underground Conductors:

- 1). Material: Bare, medium hard drawn, stranded copper.
- 2). Size: 4/0 AWG, unless otherwise shown on drawings.

G. Bare Copper Conductors: Comply with the following:

- 1). Solid Conductors: ASTM B 3 – (soft or annealed).
- 2). Assembly of Stranded Conductors : ASTM B 8 (soft, medium-hard, or hard drawn).

H. Copper Bonding Conductors: As follows:

- 1). Bonding Conductor: stranded copper conductor. Sized per code
- 2). Bonding Jumper: Bare copper tape, braided bare copper conductors, terminated with copper ferrules.

I. Ground Conductor and Conductor Protector for Wood Poles: As follows:

- 1). No. 4 AWG minimum, soft-drawn copper conductor.
- 2). Conductor Protector: Half-round PVC or wood molding. If wood, use pressure-treated fir.

J. Grounding Bus: Bare, annealed copper bars of rectangular cross section.

2.3 CONNECTOR PRODUCTS

- A. Comply with IEEE 837 (Substations) and UL 467; listed for use for specific types, sizes, and combinations of conductors and connected items.
- B. Mechanical Connectors: Bolted-pressure-type connectors (split bolt), or compression type (set screw).
- C. Welded Connectors: Exothermic-welded type, in kit form, and selected per manufacturer's written instructions.
- D. Crimp Connector: Irreversible type

2.4 GROUNDING ELECTRODES

- A. Ground Rods: Copper-clad steel.

- 1). Size: Minimum 5/8 by 96 inches (16 by 2400 mm) or as otherwise shown on drawings.
- B. Chemical Electrodes: Copper tube, straight or L-shaped, filled with non-hazardous chemical salts, terminated with a 4/0 bare conductor. Provide backfill material recommended by manufacturer.
- C. Test Wells:
 - 1). Well Pipe : Pipe shall be 8 inch diameter by 12" long concrete pipe with belled end.
 - 2). Well Cover : Cast iron with legend "GROUND" embossed on the cover.
 - 3). Manufacturer : Brooks Products- Type 3RT or equal if approved by the Owner.

PART 3 EXECUTION

3.1 GENERAL

- A. The grounding electrodes shall be bonded together to form a complete grounding electrode system per NEC. See drawings for additional details.
- B. System components shall be installed in accordance with the manufacturer's instructions.
- C. Ground non-current carrying metal parts of electrical equipment enclosures, frames, conductor raceways, and cable trays, to provide a low impedance path for possible ground fault current and to bond all non-current carrying metal parts together.
- D. Bonding shall be provided to assure electrical continuity and the capacity to conduct any fault current safely.
- E. All ground conductors shall be electrically continuous from the electrical circuit source to the equipment being grounded.

3.2 APPLICATION

- A. In raceways, use insulated equipment grounding conductors.
- B. Exothermic-Welded Connections: Use for connections to structural steel and for underground connections, except those at test wells.

- C. Equipment Grounding Conductor Terminations: Use termination appropriate for the equipment connected to.
- D. Ground Rod Clamps at Test Wells: Use bolted pressure clamps with at least two bolts.
- E. Grounding Bus: ¼" thick x 4" x 12" minimum or as shown on drawings. Install in electrical, in rooms housing service equipment, and elsewhere as indicated. See Section 17011 (Communications Grounding) for additional locations.
- F. Use insulated spacer; space 1 inch (25.4 mm) from wall and support from wall 6 inches (150 mm) above finished floor, unless otherwise indicated.
- G. Underground Grounding Conductors: Use copper conductor, No. 4/0 AWG minimum. Bury at least 24 inches (600 mm) below grade.

3.3 EQUIPMENT GROUNDING CONDUCTORS

- A. Comply with NFPA 70, Article 250, for types, sizes, and quantities of equipment grounding conductors, unless specific types, larger sizes, or more conductors than required by NFPA 70 are indicated.
- B. Install equipment grounding conductors in all feeders and circuits.
- C. Busway Supply Circuits shall include equipment grounding conductor.
- D. Isolated Grounding Receptacle Circuits: Install an insulated equipment grounding conductor connected to the receptacle grounding terminal. Isolate grounding conductor from raceway.
- E. Isolated Equipment Enclosure Circuits: For designated equipment supplied by a branch circuit or feeder, isolate equipment enclosure from supply raceway with a nonmetallic raceway fitting listed for the purpose. Install fitting where raceway enters enclosure, and install a separate equipment grounding conductor. Isolate equipment grounding conductor from raceway.
- F. Signal and Communication Systems: See Section 17011 – Communications Grounding.
- G. Common Ground Bonding with Lightning Protection System: Bond electrical power system ground directly to lightning protection system grounding conductor at closest point to electrical service grounding electrode. Use bonding conductor sized same as system grounding electrode conductor, and install in conduit.

3.4 GROUND RING (Counterpoise)

- A. Ground the steel framework of the building, see drawings. Use medium-hard drawn stranded copper conductor not less than No. 4/0 AWG for ground ring and for tap to building steel. Bury ground ring not less than 18 inches (450 mm) below grade and 24 inches (600 mm) from building foundation.

3.5 RAISED FLOOR ESD GROUNDING:

- A. A ground ring (#4/0 medium-hard drawn bare stranded copper conductor) shall be provided around every 2500 sq. ft. of raised floor. Every 5th raised floor pedestal shall be connected to the ground ring with #6 AWG stranded copper conductors. See drawings for details.
- B. Resistance shall be 1 megaohm +/- 0.5 megaohm from any floor tile to the ground ring.

3.6 INSTALLATION

- A. Ground Rods:
 - 1). Drive ground rods below finished floor or final grade.
 - 2). Locate per drawings, install per NEC.
 - 3). Interconnect ground rods with grounding electrode conductors. Use exothermic welds, except at test wells and as otherwise indicated. Make connections without exposing steel or damaging copper coating.
- B. Grounding Conductors: Avoid obstructing access or placing conductors where they may be subjected to strain, impact, or damage.
- C. Install ground rod test wells where shown on drawings. Set top of well flush with finished grade or floor. Soils shall be compacted under and around ground rod test wells.
- D. Ufer Ground (Concrete-Encased Grounding Electrode): Fabricate according to NEC. If concrete foundation is less than 20 feet (6 m) long, coil excess conductor within the base of the foundation. Bond grounding conductor to reinforcing steel. Extend grounding conductor below grade and connect to building grounding grid or to a grounding electrode external to concrete.

3.7 CONNECTIONS

- A. General: Make connections so galvanic action or electrolysis possibility is minimized. Select connectors, connection hardware, conductors, and connection methods so metals in direct contact will be galvanically compatible.
 - 1). Use electroplated or hot-tin-coated materials to ensure high conductivity and to make contact points closer to order of galvanic series.
 - 2). Make connections with clean, bare metal at points of contact.
- B. Exothermic-Welded Connections: Comply with manufacturer's written instructions.
- C. Equipment Grounding Conductor Terminations: For No. 8 AWG and larger, use pressure-type grounding lugs.
- D. Motors shall be connected to equipment ground conductors with a bolted lug connection on the metal frame. Clean surfaces and remove paint prior to connection.
- E. Non-contact Metal Raceway Terminations: If metallic raceways terminate at metal housings without mechanical and electrical connection to housing, terminate each conduit with a grounding bushing. Connect grounding bushings with a bare grounding conductor to grounding bus or terminal in housing. Bond electrically non-continuous conduits at entrances and exits with grounding bushings and bare grounding conductors, unless otherwise indicated.
- F. Connections at Test Wells: Connections to ground rods shall be an NEC approved method capable of being disconnected for testing.
- G. Tighten screws and bolts for grounding and bonding connectors and terminals according to manufacturer's published torque-tightening values. If manufacturer's torque values are not indicated, use those specified in UL 486A.
- H. Compression-Type Connections: Use hydraulic compression tools to provide correct circumferential pressure for compression connectors. Use tools and dies recommended by connector manufacturer. Provide embossing die code or other standard method to make a visible indication that a connector has been adequately compressed on grounding conductor.

3.8 OVERHEAD-LINE GROUNDING

- A. Comply with IEEE C2 (NESC) requirements. Use 2 or more parallel ground rods if a single ground rod electrode resistance to ground exceeds 25 ohms.
- B. Drive ground rods below finished grade in undisturbed earth at least 2 feet from pole.

- C. Ground Rod Connections: Use clamp-type connectors listed for the purpose for underground connections and connections to rods.
- D. Lightning Arresters: Separate arrester grounds from other grounding conductors.
- E. Secondary Neutral and Tank of Transformer: Interconnect and connect to grounding conductor.

3.9 UNDERGROUND DISTRIBUTION SYSTEM GROUNDING

- A. Install ground rods at opposite corners, outside of manholes (vaults). Install a 1/4" x 2" x 18" copper ground bar on all interior walls, or as shown on drawings, with 1" clear space behind the ground bar. Connect ground rods to ground bars with No. 4/0 bare stranded medium-hard drawn copper. Connect all ground bars with a No. 4/0 bare stranded medium-hard drawn copper ring inside the manhole. Make connection at ground bar with a service split bolt. Make connection at ground rod with exothermic weld (Cadweld) or clamped by approved means.
- B. Mechanical lugs installed on ground bus shall be manufactured from copper or brass. Drain wires from high voltage terminations shall be individually terminated, multiple wires under a common lug is not acceptable.
- C. Connections to Manhole Components: Connect exposed-metal parts, such as inserts, cable racks, pulling irons, and ladders within each manhole, to a vault ground bar or ring. Cable shields shall be connected directly to a ground bar. Make connections with No. 6AWG minimum, stranded, medium-hard drawn copper. Train conductors level or plumb around corners and fasten to manhole walls. Connect to cable armor and cable shields as recommended by manufacturer of splicing and termination kits. Make connections to ladder at top.
- D. Pad-Mounted Transformers and Switches: Install ground rods and ground ring circling pad as shown on drawings. Ground pad-mounted equipment and non-current-carrying metal items associated with substations, including fences, by connecting them to underground cable and grounding electrodes. Use medium hard drawn stranded copper conductor not less than No. 4/0 AWG for ground ring and for taps to equipment. Bury ground ring not less than 18 inches (450 mm) below grade and 6 inches (150 mm) from the foundation, 36" from fence. Provide flexible braided jumper from fence to gates.

3.10 FIELD QUALITY CONTROL

- A. All grounding system connections below grade shall be inspected by the Authority Having Jurisdiction and the Owner or his designee for mechanical integrity to insure a complete system prior to burial.
- B. Testing: Perform the following field quality-control testing:
 - 1). After installing grounding system but before permanent electrical circuitry has been energized, test for compliance with requirements.
 - 2). Test completed grounding system at service disconnect enclosure grounding terminal, and at ground test wells. Measure ground resistance not less than two full days after the last trace of precipitation, and with-out the soil being moistened by any means other than natural drainage or seepage and without chemical treatment or other artificial means of reducing natural ground resistance. Perform tests, by the fall-of-potential method according to IEEE 81.
 - 3). Provide drawings locating each ground rod and ground rod assembly and other grounding electrodes, identify each by letter in alphabetical order, and key to the record of tests and observations. Include the number of rods driven and their depth at each location and include observations of weather and other phenomena that may affect test results.
 - a). Equipment Rated 500 kVA and Less: 10 ohms
 - b). Equipment Rated 500 to 1000 kVA: 5 ohms
 - c). Equipment Rated More Than 1000 kVA: 3 ohms
 - d). Substations and Pad-Mounted Switching Equipment: 5 ohms
 - e). Manhole Grounds: 10 ohms
 - 4). Excessive Ground Resistance: If resistance to ground exceeds specified values, notify the Owner and his designee promptly and include recommendations to reduce ground resistance.

3.11 GRADING AND PLANTING

- A. Comply with Division 2 Section "Landscaping." Maintain restored surfaces. Restore disturbed paving as indicated.

END OF SECTION